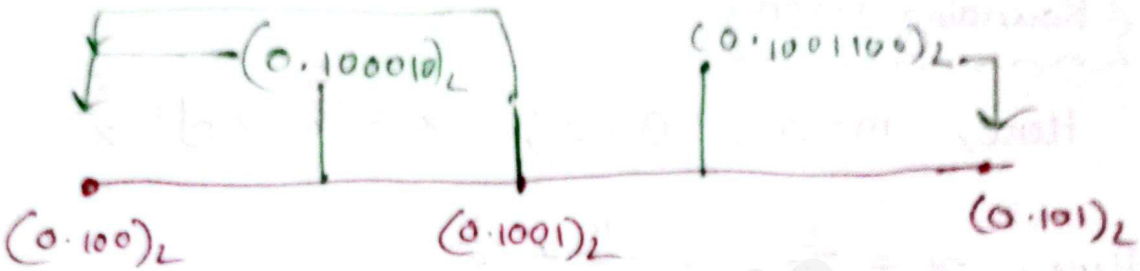
Chapter-1 [Continuation]



Rounding Error

Here, $\beta = 2$; $n = 3$; $x = \frac{5}{8}$; $y = \frac{7}{8}$

Here, $x = \frac{5}{8} = \frac{4}{8} + \frac{1}{8} = \frac{1}{2} + \frac{1}{2^3} = 2^{-1} + 2^{-3} = (0.101)_2$

$y = \frac{7}{8} = \frac{6}{8} + \frac{1}{8} = \frac{4}{8} + \frac{2}{8} + \frac{1}{8} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$

$= \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3}$

$= 2^{-1} + 2^{-2} + 2^{-3} = (0.111)_2$

Here,

$x = \frac{5}{8}$

$= \frac{4}{8} + \frac{1}{8}$

$= \frac{1}{2} + \frac{1}{2^3}$

$= 2^{-1} + 2^{-3}$

$= (0.101)_2$

$y = \frac{7}{8}$

$= \frac{6}{8} + \frac{1}{8}$

$= \frac{4}{8} + \frac{2}{8} + \frac{1}{8}$

$= \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$

$= \frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3}$

$= 2^{-1} + 2^{-2} + 2^{-3}$

$= (0.111)_2$

Rounding Error

Here, $m = 3$; $\beta = 2$; $x = \frac{5}{8}$, $y = \frac{7}{8}$

Here, $x = \frac{5}{8} = \frac{4}{8} + \frac{1}{8}$

$= \frac{1}{2} + \frac{1}{2^3}$

$= 2^{-1} + 2^{-3} = (0.101)_2$

$y = \frac{7}{8} = \frac{6}{8} + \frac{1}{8} = \frac{4}{8} + \frac{2}{8} + \frac{1}{8}$

$= \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$

$= \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3}$

$= 2^{-1} + 2^{-2} + 2^{-3} = (0.111)_2$

- Calculate $fl(x * y)$

$(x * y) = \frac{5}{8} \times \frac{7}{8} = \frac{34}{64}$

$= \frac{32}{64} + \frac{2}{64} + \frac{1}{64}$

$= \frac{1}{2} + \frac{1}{32} + \frac{1}{64}$

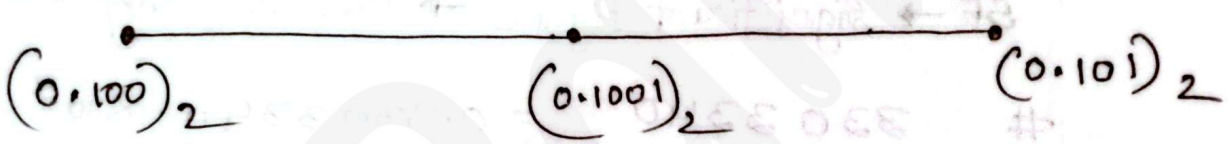
$= \frac{1}{2} + \frac{1}{2^5} + \frac{1}{2^6}$

$= 2^{-1} + 2^{-5} + 2^{-6}$

So, $(x * y) = (0.100011)_2$

- as our device [mantissa can hold 3 device so, we need to round it up.

So, $\longleftarrow \cdot (0.100011)_2$



Here, $(0.100011)_2$ which will lie in the left one

So, $(0.100011) \Rightarrow (0.100)_2$

Now, $(0.100)_2 = \frac{1}{2}$; $(0.100011)_2 = \frac{35}{64}$

$\frac{35}{64} \neq \frac{1}{2}$; So, rounding error exists.

Loss of Significance

73.98725

4SF → 73.99 [Non zero]

0.00078 3259

4SF → 0.00078833 [Non zero]

SF → Significant Figure → loss ∵ only when we subtract

3303340 = 0.3303340 $\times 10^7$

3SF → 0.330 $\times 10^7$

- when we input x, computer auto roundup to $fl(x)$

$x \rightarrow fl(x)$

$fl(x) \neq x$

~~$fl(x) = x(1+\delta)$~~

$fl(x) = x + \delta_1 x$

So, $fl(x) = x(1+\delta_1)$

$y \rightarrow fl(y)$

$fl(y) \neq y$

$fl(y) = y + \delta_1 y$

$fl(y) = y(1+\delta_1)$

So, $(x \pm y) \longrightarrow fl(x) \pm fl(y)$

$= x(1+\delta_1) \pm y(1+\delta_1)$

$= (x \pm y) \pm x\delta_1 \pm y\delta_2$

$= (x \pm y)\left(1 + \frac{x\delta_1 \pm y\delta_2}{x \pm y}\right)$

↳ scale invariant Error

Loss of significance

$(x - y) \rightarrow fl(x) - fl(y)$

$= x \quad \frac{x\delta_1 - y\delta_2}{x-y}$; if $x = y$: ∞

• When we subtract 2 numbers, if both the numbers are much closure then in case of rounding error, the denominator will be very low, approx = 0, in that case, the scale invarient error will be very high or it tends to be infinite. That will raise a huge error which is called loss of significance

Find the roots [4SF]

$x^2 - 56x + 1$

So, $x_1 = 28 + 3\sqrt{87} = 55.98$

$x_2 = 28 - 3\sqrt{87} = 0.01786$

Our computer can read upto 4SF.

[How our computer will solve it]

Computation by computer

$x_1' = 28 + 3\sqrt{87} = 28 + 27.98 = 55.98$

$x_2' = 28 - 3\sqrt{87} = 28 - 27.98 = 0.02000$

loss of significance

$ax^2 + bx + c$; $\alpha\beta = \frac{c}{a}$; $\alpha + \beta = -\frac{b}{a}$

$\alpha\beta = \frac{c}{a}$

$\therefore \beta = \frac{c}{a} / \alpha = \frac{1/1}{55.98} = 0.01786$

Relative Error $= \frac{|0.01786 - 0.02|}{0.01786} \approx 0.1198$

$\approx 11.98\%$

Practice Sheet

1. Ans

standard → $\pm (0 \cdot d_1 d_2 d_3 \cdots d_m)_\beta \times \beta^e$

Normalized → $\pm (1 \cdot d_1 d_2 d_2 \cdots d_m)_\beta \times \beta^e$

DeNormalized → $\pm (0 \cdot 1\, d_1 d_2 d_3 \cdots d_m)_\beta \times \beta^e$

(a) $\beta = 2,\ m = 4;\quad -3 \le e \le 4$

So, $e = \{-3, -2, -1, 0, 1, 2, 3, 4\}$

$= 8$

Without negative Support,

	Max (Binary)	Min (Binary)	Max (D)	Min D
standard	$(0.1111)_2 \times 2^4$	$(0.1000)_2 \times 2^{-3}$	15	0.0625
Norm	$(1.1111)_2 \times 2^4$	$(1.0000)_2 \times 2^{-3}$	31	0.125
DeNorm	$(0.11111)_2 \times 2^4$	$(0.10000)_2 \times 2^{-3}$	15.5	0.0625

Binary to Decimal

$(1.1111)_2 = (?)_{10}$

$1\times2^0 + 1\times2^{-1} + 1\times2^{-2} + 1\times2^{-3} + 1\times2^{-4}$

$= (1.9375)_{10}$

(b)

Standard: $2\times2\times2\times8$

Norm: $2\times2\times2\times2\times8$

DeNorm: $2\times2\times2\times2\times8$

(c)

Standard form

$(0.1---)_\beta \times \beta^e$; without negative suppose

	e^{-3}	e^{-2}	e^{-1}	e^0	
0.1 0 0					0 → $(0.5)_{10}$
0.1 0 0					1 → $(0.5625)_{10}$
0.1 0 1					0 → $(0.625)_{10}$
0.1 0 1					1 → $(0.6875)_{10}$
0.1 1 0					0 → $(0.75)_{10}$
0.1 1 0					1 → $(0.8125)_{10}$
0.1 1 1					0 → $(0.875)_{10}$
0.1 1 1					1 → $(0.9375)_{10}$

(c) We know,
standard form,

$$(0.1_\ _\ _)_{\beta} \times \beta^{e}$$

$0.1000 \rightarrow (0.5)_{10}$
$0.1001 \rightarrow (0.5625)_{10}$
$0.1010 \rightarrow (0.625)_{10}$
$0.1011 \rightarrow (0.6875)_{10}$
$0.1100 \rightarrow (0.75)_{10}$
$0.1101 \rightarrow (0.8125)_{10}$
$0.1110 \rightarrow (0.875)_{10}$
$0.1111 \rightarrow (0.9375)_{10}$

mantissa	2^{-3}	2^{-2}	2^{-1}	2^{0}	2^{1}	2^{2}	2^{3}	2^{4}
0.5	0.0625	0.125	0.25	0.5	1.0	2.0	4.0	8.0
0.5625	0.0703	0.1406	0.2812	0.5625	1.125	2.25	4.5	9.0
0.625	0.0781	0.1562	0.3125	0.625	1.25	2.5	5.0	10.0
0.6875	0.0859	0.1719	0.3437	0.6875	1.375	2.75	5.5	11.0
0.75	0.0937	0.1875	0.375	0.75	1.5	3.0	6.0	12.0
0.8125	0.1016	0.2031	0.4062	0.8125	1.625	3.25	6.5	13.0
0.875	0.1094	0.2188	0.4375	0.875	1.75	3.5	7.0	14.0
0.9375	0.1172	0.2344	0.4687	0.9375	1.875	3.75	7.5	15.0

(d)

smallest Positive number,

$$(0.1000\ldots)_2 \times 2^{-1022}$$

largest positive number

$$(0.1111\ldots)_2 \times 2^{1023}$$

Here, 1 bit sign (s), 11 bit e, 52 bit m

The general formula for a floating-point number is,

$$F = (-1)^S \times 1.M \times 2^{(E-1023)}$$

Since E = 0; and m = 0; 0 - 1022 [small]

E = 2046; and m = 0; 2046 - 1023 [large]

(2047 is reserved for infinity)

(e) Effect of Exponent Bias change,

(Bias = 5w)

New smallest val → $(0.1000\ldots)_2 \times 2^{-5w}$

New ~~largest~~ val → $(0.111\ldots)_2 \times 2^{5w}$

2

$x = \frac{3}{8} = \frac{2}{8} + \frac{1}{8} = \frac{1}{4} + \frac{1}{8}$

$= 2^{-2} + 2^{-3} = (0.011)_2 = (0.1100)_2 \times 2^{-1}$

$y = \frac{5}{8} = \frac{4}{8} + \frac{1}{8} = \frac{1}{2} + \frac{1}{8}$

$= 2^{-1} + 2^{-3} = (0.101)_2 = (0.1010)_2$

So, $(x * y) = \frac{3}{8} \times \frac{5}{8} = \frac{15}{64}$

$= \frac{14}{64} + \frac{1}{64}$

$= \frac{8}{64} + \frac{6}{64} + \frac{1}{64}$

$= \frac{1}{8} + \frac{4}{64} + \frac{2}{64} + \frac{1}{64}$

$= \frac{1}{2^3} + \frac{1}{2^4} + \frac{1}{2^5} + \frac{1}{2^6}$

$= 2^{-3} + 2^{-4} + 2^{-5} + 2^{-6} = (0.001111)_2$

$= (0.1111)_2 \times 2^{-2}$

So, $(x * y) = (0.001111)_2 = (0.1111)_2 \times 2^{-2}$

$fl(x * y) = (0.1111)_2 \times 2^{-2}$ ∴ No rounding error as, $(x * y) = fl(x * y)$

3. Ans [6 SF]

$$x^2 - 60x + 1 = 0$$

So, $x_1 = 30 + \sqrt{899} = 59.983$

$x_2 = 30 - \sqrt{899} = 0.016687$

For reading upto 6 SF.

$30 + 29.9833 = 59.9833$

$x_1' = 30 + \sqrt{899} = 30 + 29.98 = 59.983$

$x_2' = 30 - \sqrt{899} = 30 - 29.98 = 0.0167$ ~~0.002~~

$x_1' = 30 + \sqrt{899} = fl(30) + fl(\sqrt{899})$

$= 59.983$

$x_2' = 30 - \sqrt{899} = fl(30) - fl(\sqrt{899})$

$= 0.02$

to compute x_2' more accurately,

$ax^2 + bx + c$; $\alpha\beta = \frac{c}{a}$

~~$x_2' = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$~~ $\alpha = x_1$, $\beta = x_2$

So, $\alpha\beta = \frac{c}{a}$

$\therefore \beta(x_2) = \frac{c/a}{\alpha} = \frac{1/1}{59.983} = 0.016677$

(Ans)

Consider the quadratic equation $x^2 - 60x + 1 = 0$. Working to 6 SF, compute the roots of the quadratic equation and check that there is a loss of significance. Find the correct roots such that loss of significance does not occur.

We know,

$$\text{Discriminant} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

So, discriminant $= \dfrac{-(-60) \pm \sqrt{60^2 - 4 \cdot 1 \cdot 1}}{2 \cdot 1}$

$= 30 \pm \sqrt{895}$

$= 30 \pm 29.9833$

So, $x_1 = 59.9833$; $x_2 = 0.0166778$

$= 0.0166700$

But the product of the root must be equal to 1. but here we get = 5. 0.0166712 / 0.0166700

Now, computation by computer

x_1' $\quad \alpha\beta = \frac{c}{a}$

$\Rightarrow \beta = \frac{c/a}{\alpha} = \frac{1/1}{59.3831} = 0.0166813$

8 $x^2 - 12x + 5$

(a) Compute the roots of the quadratic eq [4SP].

(b) Explain how loss of significance occurs in this case due to the subtraction of nearly equal numbers.

(c) Now convert back to decimal form the stored vals you obtained from the previous part, and calculate the rounding errors of both numbers.

Soln

(a)

We know, discriminant = $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

So, $= \frac{-(-12) \pm \sqrt{(-12)^2 - 4(5)(1)}}{2(1)}$

So, $\alpha = 6 + \sqrt{31} = 11.57$

$\beta = 6 - \sqrt{31} = 0.4322$

Now, computation by computer,

$\alpha' = 6 + 5.567 = 11.57$

$\beta' = 6 - 5.567 = 0.4330$

So, root,

$\alpha\beta = \frac{c}{a}$

So, $\beta = \frac{c/a}{\alpha}$

$= \frac{5/1}{11.57}$

$= 0.4322$